

Jonas Ngnawé

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Education

Université Laval, Mila - Quebec AI Institute, Stanford University (VSR)

Quebec City, Canada

PhD in Computer Science: Adversarial Machine Learning

2021–2027 (expected)

AIMS African Master's of Machine Intelligence

Kigali, Rwanda

MSc in Machine Learning

2018–2019

Funded by Meta and Google; taught by world top researchers.

African Institute for Mathematical Sciences (AIMS)

Limbe, Cameroon

MSc in Mathematical Sciences

2016–2017

École Polytechnique Yaoundé

Yaoundé, Cameroon

Engineering Degree (MSc equivalent), Computer Engineering

2011–2016

Five-year integrated engineering programme.

Summer Schools: Machine Learning Summer School (MLSS), London 2019; Deep Learning & Reinforcement Learning Summer School (DLRL), Edmonton 2025.

Publications

Peer-reviewed conference papers.....

- Detecting Brittle Decisions for Free: Leveraging Margin Consistency in Deep Robust Classifiers. **Ngnawé, J., Sahoo, S., Pequignot, Y., Précioso, F., & Gagné, C. NeurIPS 2024.** [Proceedings]
- Robust Fine-Tuning from Non-Robust Pretrained Models: Mitigating Suboptimal Transfer with Epsilon-Scheduling. **Ngnawé, J., Heuillet, M., Sahoo, S., Pequignot, Y., Ahmad, O., Durand, A., Précioso, F., & Gagné, C. ICLR 2026** (earlier version: NeurIPS 2025 Workshop on Reliable ML). [OpenReview]
- A Layer Selection Approach to Test Time Adaptation. **Sahoo, S., ElAraby, M., Ngnawé, J., Pequignot, Y., Précioso, F., & Gagné, C. AAAI 2025.** [Proceedings]

Workshop papers.....

- A Guide to Robust Generalization: The Impact of Architecture, Pre-training, and Optimization Strategy. **Heuillet, M., Bhagwatkar, R., Ngnawé, J., Pequignot, Y., Larouche, A., Gagné, C., Rish, I., Ahmad, O., & Durand, A. NeurIPS 2025 Workshop on Reliable ML.** [arXiv]
- Robustmix: Improving Robustness by Regularizing the Frequency Bias of Deep Networks. **Ngnawé, J., Njifon, M. A., Heek, J., & Dauphin, Y. NeurIPS 2022 Workshop on Distribution Shifts.** [OpenReview]

Under review.....

- RobustGenBench: A Benchmark for Robust Generalization to Adversarial and Common Perturbations, with Applications to Vision and Vision-Enabled Large Language Models. **Heuillet, M., Ngnawé, J., Bhagwatkar, R., Pequignot, Y., Larouche, A., Gagné, C., Rish, I., Ahmad, O., & Durand, A.**

Master's projects.....

- Scalable Inference with Distributed Gaussian Processes using the Wasserstein Barycenter. **Ngnawé, J., Deisenroth, M.** [GitHub]
- k-Clustering Problem: Application to a Field Technician Allocation Optimization Problem. **Ngnawé, J., Zacko, E., Wandji, N., & Ndjinga, M.** [ResearchGate]

AI/ML Work Experience

Vector Institute (FastLane MLA Program)

Toronto, Canada

Machine Learning Associate

May – Aug 2026

- Optimized LLM inference for AutoInference.ai: vLLM serving config to increase goodput and reducing latency.

Stanford Artificial Intelligence Lab (SAIL)

Visiting Student Researcher

Palo Alto, USA

Oct 2025 – June 2026

- Safety, robustness and uncertainty estimation in multimodal vision–language models.

Google Brain

Google AI Resident

Accra, Ghana

Oct 2019 – Sep 2021

- Developed robust training methods for deep networks; frequency-bias regularization work published at the NeurIPS 2022 Distribution Shifts workshop.
- Compressed deep networks via quantization and pruning: Mixed-precision quantization (4, 8-bit quantization, with 0-bit for sparsity to retain high accuracy on vision models).

Earlier Software Engineering Experience

GroupOne Holding Company

Computer Engineer

Limbe, Cameroon

Jan – Aug 2018

Web development and fuel-consumption prediction; industrial maintenance of Honeywell Experion refinery systems. *JavaScript, PHP, MySQL*.

Freelance

Software Developer

Cameroon

2017–2018

Digitized administrative procedures for Cameroonian ministries (Maarch RM electronic document management); built and maintained WooCommerce plugins. *PHP, Java, MySQL, Git*.

Bolloré Africa Logistics

Software Developer Intern

Douala, Cameroon

Feb – Aug 2016

Automated review of information-system access rights and built anomaly detection for abnormal privileges. *Talend, QlikView, C#, ASP.NET, SQL Server*.

Talks

Host, Kmer AI Talks.

2024: Panelist, “AI Regulation & Fairness in the Generative AI Era,” Black in AI Workshop @ NeurIPS.

2024–2025: “Detecting Brittle Decisions for Free” — Kmer AI (Jun 2024); Galsen AI (Jan 2025).

2022: Keynote, “Robustness in AI,” Deep Learning IndabaX-Cameroon (Dec 2022).

Service

Reviewer: NeurIPS, ICLR, AAAI; IEEE Transactions on Image Processing.

Algorithmic Fairness workshops @ NeurIPS 2021–2024; Montreal AI Symposium (2022, 2024); Black in AI Workshop (2022–2024); Deep Learning Indaba / Pan-African ML Conference (2023).

Organizing: Interpolate Workshop @ NeurIPS 2022 (First Workshop on Interpolation Regularizers and Beyond), organizing committee.

Co-organizer, Machine Learning Projects Competition (MLPC) 2020–2022, Polytechnique Yaoundé, \$2,000 sponsorship.

Co-organizer: two-week introductory ML course (Sep 2019), Polytechnique Yaoundé; first Deep Learning IndabaX Rwanda (Apr 2019), Kigali.

Co-founder, Kmer-AI — association promoting AI and ML in Cameroon (since 2020).

AI/ML Skills

Frameworks: PyTorch, TensorFlow, Scikit-Learn

Infrastructure: Distributed training (Slurm, multi-GPU), vLLM, Alperf, Docker, Weights & Biases, Google Cloud Platform, Git

Research Related: Adversarial robustness, training large VLMs from scratch, robust evaluation of LLMs/LVLMs, uncertainty estimation